

BORDER OPTICS

A Satellite-Based Verification Framework for India's Vibrant Villages Programme (VVP-I)

Executive Summary · DOI: 10.5281/zenodo.21759970 · Sakshi D. Maske

Project Overview

The origins of BORDER OPTICS lie in one line in a Parliamentary reply. Responding directly to the question, MHA said that no impact assessment has ever been conducted for the Vibrant Villages Programme, which already approved unlikely investment of Rs. something like that. The amount of Rs. 4800 Cr spread across 2967 villages, covering five Himalayan border states and UT's from Arunachal Pradesh to Ladakh. There's an intersection of three fields that I don't want to consider separately—the gap that I'm digging to fill. It only needs to be trailed by political science, given that VVP is a textbook example of the kind of infrastructure that does double duty as an outbreak of military hardware, in this case a bridge, as both civilian welfare and geopolitical signaling. The most common reason for this is economics, a large-scale public investment which has never been costed against the results it delivers. In part, because it allows, by means of satellite imagery, the Government's implicit claim of development to be directly tested instead of taken on faith, and in part because satellite imagery is an important part of geospatial science. So, I tested it. 258 villages, individually geocoded samples, randomly collected from the land and a matched non-VVP control group of 753 villages, in the same districts; equally defensible ways of compositing the same satellite record. There wasn't a simple yes or no that returned. That's one of the measurement options, and quite an important one, too; it represents a real, significant growth in built-up area, or, a completely different option, no growth at all—just a kind of instability that this project discovered in the thing, which is also, in itself, the major thing it discovered, this kind of ambiguity, really, an impact assessment must have solved, as a prerequisite, before Rs. A sum of 4,800 crore were lost. That is, wherever the surviving product could be pressure-tested -- against the control product, through a third year, through the three buffer sizes -- it stood. If cost translated to results, that would imply 10 times as much money taken to 10 times the state would expect to get. No one had been looking at Parliament, it said. This is what a Google Search reveals.

The Question

When directly asked in a Lok Sabha Unstarred Question (Lok Sabha Question No. 508) on whether any impact assessment had been done of the impact of the Vibrant Villages Programme, the Ministry of Home Affairs has responded, “No”. The question that this project aimed to answer can be traced back to this gap: Did India's government achieve its goal of reducing unemployment by introducing a mandate for India's citizens to work? What about the measurable ground level physical impact of 4,800 crore worth of development schemes in the border villages...can there be any certainty in the answer or is it a choice of whether to proceed by a different measure?

The Method

The 14 geographically matched districts included 258 villages that were individually geocoded, and 753 villages that did not, from 5 Himalayan border states and UTs, for this test. The satellite record was provided by sentinel-2 built-up-area change (NDBI) and VIIRS night-lights 2021–2025. In each case, two independent compositing windows were used, specifically full year and summer-matched. The before/after change was addressed by Wilcoxon signed-ranked tests, the budget/border proximity was addressed by Spearman correlations, and the

comparison with the control group was addressed using a district fixed-effects Difference-in-Differences model. Two more checks are added to the core design, the 2021/2022 comparison is expanded to a 3-point 2021/2022/2023 trend and the fixed extraction buffer of 500m is compared to alternative extraction buffers of 250m and 1km.

The Finding

Finally, the built-up-area result can be summarily said to be compositing-window sensitive and not confirmed. There is no significant change seen in a full-year composite, but there is a very significant change in the same villages (recomposed to a window of 1-2 months during the summer months). The dichotomy dates back to the type of artifact present on either side: snow-cover contamination in the full-year range, and monsoon cloud cover in the summer-only range, effectively eliminating Sikkim's data. Three additional stress tests were applied to this significant result in the summer and showed it to be robust, surviving all three that were applied, and also surviving multiple testing (500m Holm-Bonferroni corrected) as well as being sample-matched to the change seen in a similarly matched district-FE non-VVP control group ($p = 0.0021$). It isn't an ascending trend, though, because a 2023 to 2025 triple streak reveals the bulk of the change in a "recovery" covering 2023.

Test	Full-Year Window	Summer-Matched Window
NDBI change (Wilcoxon)	$p = 1.000$ — not significant	$p < 0.000001$ — highly significant
Night-lights change (Wilcoxon)	$p = 0.050$	$p = 0.9999$
Control-group DiD (NDBI, H4)	$p = 0.215$ — not significant	$p = 0.0021$ — significant
Budget vs. measured change	~10x budget gap (AP vs. UK) → nearly identical mean change (same in both windows)	~10x budget gap (AP vs. UK) → nearly identical mean change (same in both windows)

It is also more important as a proxy here than the headline NDBI indicator for the opposite reason, it is the consistent, more reliable measure that is not affected by that seasonal confounding factor that breaks up the NDBI. Budget also exhibits its own narrative: The sanctioned budget of Arunachal Pradesh is Rs. At the same time, its 2,749.74 crore come close to 10-times the income of Uttarakhand (Rs. Despite these disparities in size, the two states' average measured built up change values is practically indistinguishable descriptively, and compare well with what would be expected to occur if implementation is independent of budget quantity.

Validation & Robustness Checklist

- Long-term compositing window test can account for the effects of the summer months. The long-term (composite window test + summer matched) ✓ Two individual metrics tested independently (NDBI + VIIRS)
- ✓ Non-VVP control group (753 villages, 14 same districts, DiD design) Found this it didn't require three before/after comparisons, it required three years of trends (2021/2023/2025)
 - ✓ Water boundary buffer (250m/500m/1km) sensitive sweeping Cross checked against sanctioned budget (Parliamentary record) ✓ Public Interest Litigation Cases brought forth by citizens for "disclosing every data gap" (Ladakh, HP and caveats in Uttarakhand) Either the result expressed as a window-sensitive NDBI, or once significant, a window-sensitive or only a significant number of NDBIs were found between 2023 and 2025.

Honest Limitation

If this result is actually a clean confirmation, it's not the headline of this project. This is not a clean confirmation, this is an instability disclosure. The Himalayas provide their own seasonally introduced measurement artifacts: snow cover in winter months means no data at all for Sikkim; monsoon cloud cover means no data we can use in summer months. Measurement artifacts can be introduced due to the seasonal nature of the Himalayas and those artifacts are sufficient to make a single compositing choice an unreliable evidence. Comparing to the control villages enhances the remaining signal, but the comparison itself is based on an imbalance of the NDBI level before treatment with that of the control group, without any actual proof of a common pre-trend. Due to some limitations on sample size and geometry, the "border-proximity" correlation (H3) is considered a confirmatory instead of exploratory. The data gaps – both “un-solved” villages in Ladakh and “partial” samples in HP and “block” confusion in Uttarakhand – are not glossed over here.

Real-World Relevance

Parliament learnt that there was no independent impact study done in the matter of a Rs. The government scheme has been designed in the regions of 4,800 villages – and that is this project. It demonstrates the feasibility and the necessity of satellite methods for verifying public infrastructure projects and it demonstrates something else: when verifying such projects in space by using two of the most defensible measurement options yields opposite results, the useful answer for the policy maker is not to either side-steak or overlook the doubts casted on that measurement, but to examine both independently. It's revealing, first to see if there's any instability, and then stress-testing the one that stands up to a control group and a multi-year sweep and sensitivity analysis of the methodology.

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GitHub: github.com/sakshimaske303-commits/BORDER_OPTICS | Live Dashboard: borderoptics-bkx3lpcvfghdpa2hmuqwsq.streamlit.app | Zenodo DOI: [10.5281/zenodo.21759970](https://doi.org/10.5281/zenodo.21759970)

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